

NKU HNR 311 Global Citizens in the Natural World (Spring 2020)

The Extremes of Language

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Office Hours: Tuesdays 8-10 and Fridays 10:30-11:30 (or by appointment)

Seminar Meetings: Mondays and Wednesdays, 2:00-3:15, 270 Founders Hall

What are the limits of human language, and what is beyond?

בבל

We begin with a close reading of the story of Babel because it has shaped European-influenced thinking on the diversity of languages, and because (for use later in the course) it connects to a certain kind of linguistic mysticism (*kabbalah*).

မြန်မာ

Taking English as the arbitrary zero reference point (only because it is the language in which we are conducting this seminar), we explore three increasingly non-English-like languages: **Burmese** (a Tibeto-Burman language spoken in Myanmar), **Mam** (a Mayan language from Guatemala), and **Archi** (a language spoken in the Caucasus). There are many things to study in a language, and we pick two: the *sounds* of the language, and the way it recognizes (or does not recognize) the *subject* of a sentence.

hizkuntza

The latter suggests that language shapes how we think about the world, and this relates to the so-called Sapir-Whorf Hypothesis. To provoke discussion on that, we read a short science fiction novel, *Babel 17*, by Samuel R. Delany (a gay male African American writing as an Asian female linguist from the future...).

$\aleph_0$

The *kabbalah*, part of a millennia-long tradition of mysticism around the Hebrew language, lets us explore other ways humans have used language. It also, by its influence on a school of thought called Lullism, leads us to (simultaneously) talk about infinity and the origins of computer languages.

$(\lambda \ (x) \ f)$

From there it is an easy transition to learning to write some code in the language **Racket**, an especially elegant minimalist programming language whose ancestors and descendants are very influential in logic-based approaches to artificial intelligence. We end with a precise analysis of what cannot be expressed in programming languages.

$e^{-i\phi} |\uparrow\rangle$

Finally, we look to quantum physics to explore areas where logic and language *as we know them* are said to break down. We examine popular New Age language around quantum mysticism and pick out the strands of genuine paradox and mystery that truly arise from the science.

Catalog Description and Declared Learning Outcomes

From the university catalog:

HNR 311 Global Citizen in the Natural World - NS/QR (3 credits)

Hours: 3 classroom + 0 lab/studio

Prerequisite(s): HNR 210 [*Scholarly Approaches to the Natural World*]

Students will learn about the privileges and responsibilities individuals have as contributing members of a global community. These include understanding one's role as an ethical scientist, cultivating empathy for those who differ from our own, and foster awareness of one's broader impact on other people and the environment. Note: this course counts toward both NS and QR general education categories.

Repeatable: No

Practically, the prerequisite is experience with the scientific stance toward the natural world (HNR 210). Generally, as with any Honors seminar, other prerequisites are curiosity and wide-ranging interest in the broader experience of humanity.

NKU associates with this course two "Foundations of Knowledge" learning outcomes. I've provided {*annotations*} on how we work them into this seminar.

- D2. Students identify major concepts of science behind technological innovations or applications in our daily lives. {*Science: quantum mechanics; technological innovation: programming languages.*}
- D3. Students distinguish between scientific and non-science explanations by employing scientific methods. {*Scientific and non-science explanations: quantum physics/metaphysics; concepts of infinity in mathematics and elsewhere.*}

The Honors College associates three additional outcomes:

- VI. Students will recognize that global issues are complex. {*Language complexity, and attempts at preservation and restoration.*}
- IX. Students will define how personal decisions impact the greater world {*Implications of a world built on code: fragile, hackable.*}
- X. Students will demonstrate how to be a respectful global citizen. {*Respecting the diversity of human language as part of human culture.*}

Coursework: *Contribution Toward Grade* [revised]

PRE-SPRING BREAK:

Seminar interaction (and "citizenship"): 16%

3 assignments (mostly 2-3pp essays, sometimes exercises/problems): 24%

POST-SPRING BREAK:

Online discussion participation (by discussion boards on canvas or in-person synchronous) 25%

4 assignments (mostly 2-3pp essays, sometimes exercises/problems): 35%

The Seminar Experience [revised]

Dean Buss has provided some good descriptions of what proper “citizenship” at a seminar means. The descriptions include mention of:

- Politeness and respect
- Preparation
- Being on time
- Making thoughtful and thought-provoking contributions
- Speaking to everyone in the room

All these are essential. **For the second, online, portion of the course, bullets 1,2 and 4 become paramount. Above all, you want to make sure you are truly contributing, with your contributions reflecting all the time you put in. But everyone in the class is going through very challenging times; we will all try to be understanding.**

In this course, we will tackle some difficult ideas and difficult readings. Our first goal every time we meet is to work together to make sure everyone in the room understands the readings and new concepts. Expect to arrive in class with questions, and with a willingness to help answer questions from others.

In many seminar situations, a good deal of time is spent offering opinions and passing judgment. That is important here too, but secondary. Nevertheless, a keen skeptical eye is valuable.

Grade Calculation

Per university practice, I will provide you with a midterm grade, broken down by the coursework categories on the previous page. I will grade and return all assignments to you within one week of submission.

```
(define percent-to-grade
  (λ (pct)
    (cond
      ((>= pct 93) 'A) ((>= pct 90) 'A-) ((>= pct 87) 'B+) ((>= pct 83) 'B) ((>= pct 80) 'B-) ((>= pct 77) 'C+)
      ((>= pct 73) 'C) ((>= pct 70) 'C-) ((>= pct 67) 'D+) ((>= pct 63) 'D) (else 'F))))
```

Effort / Credit Hour Policy Statement

According to the NKU catalog, a 3-credit hour course requires 8-9 hours of work per week. In a traditional class, this would be 3 hours of seminar time and 5-6 hours outside class each week. Please make sure that your school/work/life schedule permits this time investment. **In online mode, it is still 8-9 hours total per week.**

Contacting the Instructor [revised]

Email is usually best, if you cannot come by my office in person. You can expect me to reply within 24 hours, or 48 hours on the weekend. We can also connect by phone **or Zoom**.

Exceptional Circumstances [revised]

You can turn in up to one assignment up to 48 hours late, although the grade will be dropped by one full letter.

Attendance is required **during the face-to-face first half of the semester**. If you absolutely cannot make a class, please contact me in advance and we will try to make exceptional arrangements.

Required Readings

Most of the readings in this course are short articles, book chapters, Internet resources, and tutorials/notes by the instructor. These are available online through Steely Library reserves and the course Canvas site.

The two exceptions are these books, which you should acquire a copies of:

Samuel R. Delany, *Babel 17*.

The book was first published in 1966 but the edition that's probably easiest is the Vintage edition from 2002 (which restores one censored word from 1966).

B. Rosenblum & F. Kuttner, *Quantum Enigma: Physics Encounters Consciousness*. 2nd Edition. Oxford University Press, 2011. (Paperback).

A short book based on a course at Santa Cruz for liberal arts majors.

We will also watch Darron Aronofsky's film π . (Note: it is rated R for "disturbing images".)

Required Software

We will write code in the *Racket* programming language. You can download the tools you need from racket-lang.org. It is free, and runs on PCs or Macs. You can wait until late February to do this. I will also make sure it is installed on some open labs on campus.

Note: We are picking the *Racket* language because it is a type of language (a dialect of *Lisp*) that is extremely influential and yet not usually studied by computer science majors. In that sense, it levels the playing field. However, if you have experience in computer programming, you may wish to use the language *Clojure* instead. It is another dialect of *Lisp* essentially identical to *Racket*, but runs on the Java Virtual Machine (which will make it more practical if you want to do real-world applications).

Schedule of Topics [revised]

This is our roadmap. The actual timing is subject to change. Details, readings, and assignments will be posted on the course Canvas site as the semester proceeds.

		TOPICS	DUE DATES
1	M Jan 13	Overview and introductions	
	W Jan 15	Some language origin stories: Eden and Babel	
2	M Jan 20	<i>University Closed: MLK Day</i>	
	W Jan 22	<i>Confusio linguarum</i> and language families	
3	M Jan 27	The sounds of language: the International Phonetic Alphabet	
	W Jan 29	The grammar of subjects and objects	Assignment 1
4	M Feb 3	Endangered languages: overview	
	W Feb 5	Interviewing native speakers: Amharic	
5	M Feb 10	A Mayan language: Mam	
	W Feb 12	Ergativity: Examples from Guugu Yimidhirr and Ho	
6	M Feb 17	Language renewal: Myaamia	
	W Feb 19	Interviewing a speaker: Myaamia	
7	M Feb 24	Interviewing native speakers: Burmese	Assignment 2
	W Feb 26	<i>Babel-17</i>	
8	M Mar 2	<i>No class</i>	
	W Mar 4	<i>Babel-17</i> and Linguistic Relativity	Assignment 3
-	M Mar 9	<i>Spring Break</i>	
	W Mar 11		
9	M Mar 16	<i>Extended Spring Break</i>	
	W Mar 18		
10	M Mar 23	Looking back (a review)	
	W Mar 25	The early Kabbalah	
11	M Mar 30	Borges Library of Babel	Assignment 4
	W Apr 1	Leibniz and blind thought	
12	Apr 6-12	Writing in Racket: Meaningless symbol manipulation is good	
13	Apr 13-19	Writing in Racket: Codecraft and the Limits of Code	Assignment 5 Due 11pm Sun Apr 19
14	Apr 20-26	Quantum Physics: The breakdown of common sense I The moon is not there if nobody looks	
15	Apr 27 - May 3	Quantum Physics: The breakdown of common sense II Consciousness	Assignment 6 Due 11pm Sun May 3
16	May 4 - 8 [Finals Week]	Self-Reference and Paradox: The breakdown of language The halting problem and the exploding computer	Assignment 7 Due 11pm Fri May 8